

Customer Shopping Behavior Analysis

1. Project Overview

This project analyses customer shopping behavior using transactional data from 3900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic behavior decisions.

2. Dataset Summary

-Rows:3900

-Columns:18

-Key Features:

- Customer demographic (Age, Gender, Location, Subscription status)
- Purchase Det (Item purchase,, Category Purchased amount Season, size, colour)
- Shopping behaviour (Discount applied, Promo Code used, Previous purchases, Frequency of purchases, Review rating, Shipping type)
- Missing data: 37 values in review rating column

3. Exploratory Data Analysis using Python

We begin with data preparation and cleaning in python:

- Data Loading: Imported the dataset using pandas.
- Initial Exploration: Used `df.info()` to check structure and `describe()` for Summary statistics.

Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type
2658	52	Female	Dress	Clothing		81 Texas	M	Pink	Summer	2.7	No	Express
2663	25	Female	Skirt	Clothing		98 Nebraska	M	Cyan	Summer	2.8	No	2-Day Shipping
2674	67	Female	Hoodie	Clothing		21 Nevada	M	Olive	Summer	2.6	No	2-Day Shipping
2676	24	Female	Shorts	Clothing		38 Virginia	M	Peach	Winter	3.4	No	Free Shipping
2685	46	Female	Dress	Clothing		45 Washington	M	Gray	Summer	4.7	No	Free Shipping
2687	40	Female	Socks	Clothing		72 Georgia	M	Black	Winter	3.8	No	2-Day Shipping
2688	61	Female	Dress	Clothing		75 Delaware	M	White	Fall	3.6	No	2-Day Shipping
2690	29	Female	Blouse	Clothing		35 Wyoming	M	Orange	Fall	3.9	No	Standard
2699	55	Female	Hoodie	Clothing		57 Virginia	M	Magenta	Summer	3.6	No	Store Pickup
2709	57	Female	Dress	Clothing		94 Alabama	M	Teal	Summer	3.3	No	Store Pickup
2719	22	Female	Shorts	Clothing		90 Virginia	M	Peach	Winter	4.9	No	Next Day Air
2729	26	Female	Skirt	Clothing		28 Florida	M	Pink	Winter	4	No	Express
2736	46	Female	Skirt	Clothing		29 California	M	Yellow	Spring	3	No	Express
2737	44	Female	Hoodie	Clothing		25 Montana	M	Pink	Spring	4.8	No	Store Pickup
2745	38	Female	Shirt	Clothing		66 Wyoming	M	Silver	Winter	3.3	No	Store Pickup
2747	39	Female	Hoodie	Clothing		25 Ohio	M	Orange	Fall	3.7	No	Standard
2748	18	Female	Jeans	Clothing		62 West Virginia	M	Teal	Winter	3.1	No	Free Shipping
2759	32	Female	Blouse	Clothing		98 Illinois	M	Purple	Summer	3.2	No	Standard
2763	58	Female	T-shirt	Clothing		41 Georgia	M	Charcoal	Fall	3.8	No	Store Pickup
2767	66	Female	Dress	Clothing		91 Tennessee	M	Cyan	Winter	2.8	No	Standard
2768	31	Female	Dress	Clothing		95 Iowa	M	Green	Spring	4.5	No	Free Shipping
2769	55	Female	Blouse	Clothing		25 Kansas	M	Beige	Spring	3.8	No	2-Day Shipping
2778	52	Female	Blouse	Clothing		24 Idaho	M	Magenta	Fall	4.8	No	Standard
2784	19	Female	Skirt	Clothing		38 Virginia	M	Silver	Spring	4.5	No	Next Day Air
2785	33	Female	Skirt	Clothing		70 Idaho	M	Black	Winter	3.1	No	Express

Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases	Age_group	purchase_frequency_days
Express	No	No	7	Credit Card	Fortnightly	Middle-aged	Fortnightly
2-Day Shipping	No	No	44	PayPal	Fortnightly	Young Adult	Fortnightly
2-Day Shipping	No	No	4	Cash	Monthly	Senior	Monthly
Free Shipping	No	No	38	Cash	Bi-Weekly	Young Adult	Bi-Weekly
Free Shipping	No	No	42	Credit Card	Bi-Weekly	Middle-aged	Bi-Weekly
2-Day Shipping	No	No	33	Cash	Fortnightly	Adult	Fortnightly
2-Day Shipping	No	No	12	Debit Card	Bi-Weekly	Senior	Bi-Weekly
Standard	No	No	47	PayPal	Quarterly	Young Adult	Quarterly
Store Pickup	No	No	34	Venmo	Quarterly	Middle-aged	Quarterly
Store Pickup	No	No	44	Debit Card	Weekly	Middle-aged	Weekly
Next Day Air	No	No	10	Venmo	Bi-Weekly	Young Adult	Bi-Weekly
Express	No	No	17	Debit Card	Weekly	Young Adult	Weekly
Express	No	No	2	PayPal	Bi-Weekly	Middle-aged	Bi-Weekly
Store Pickup	No	No	45	Credit Card	Annually	Adult	Annually
Store Pickup	No	No	40	Credit Card	Monthly	Adult	Monthly
Standard	No	No	21	PayPal	Every 3 Months	Adult	Every 3 Months
Free Shipping	No	No	21	Credit Card	Every 3 Months	Young Adult	Every 3 Months
Standard	No	No	8	Bank Transfer	Bi-Weekly	Adult	Bi-Weekly
Store Pickup	No	No	10	Debit Card	Every 3 Months	Senior	Every 3 Months
Standard	No	No	47	Credit Card	Monthly	Senior	Monthly
Free Shipping	No	No	31	Cash	Monthly	Young Adult	Monthly
2-Day Shipping	No	No	28	PayPal	Bi-Weekly	Middle-aged	Bi-Weekly
Standard	No	No	28	Credit Card	Every 3 Months	Middle-aged	Every 3 Months
Next Day Air	No	No	26	Credit Card	Quarterly	Young Adult	Quarterly
Express	No	No	12	PayPal	Every 3 Months	Adult	Every 3 Months

. Missing data handling Cheque for null values and inputted missing values in Review rating column using the median rating of each product category.

.Column standardization: Renamed it column to snake case or better credibility and documentation.

Feature Engineering:

- Created age_group columns by binding customer ages.
- Created purchase_frequency_days from purchases day.

. Data consistency cheque verified if discount_applied and promo code used were redundant dropped promo _code_used.

4.Data Analysis using SQL

We performed structured analysis in PostgreSQL to answer key features questions:

	Gender text	revenue numeric
1	Female	75191
2	Male	157890

2.High-Spending Discount Users—Identified customers who used discounts but still spent above the average purchase amount.

	Customer ID bigint	Purchase Amount (USD) bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97

3.Top 5 products by Rating – Found products with the highest average review ratings.

	Item Purchased text	Average Product Rating double precision
1	Gloves	3.8627737226277383
2	Sandals	3.8446540880503144
3	Boots	3.818881118881119
4	Hat	3.801307189542483
5	Skirt	3.7853503184713366

4. Compare the average purchased amount between standard and express shipping?

	Shipping Type text	Average Purchase Amount numeric
1	Standard	58.4602446483180428
2	Express	60.4752321981424149

5. Do subscriber customer spend more compare average spend and total revenue between subscribers and non subscribers?

	Subscription Status text	Average Purchase Amount numeric	Total Revenue numeric
1	No	59.8651211801896733	170436
2	Yes	59.4919278252611586	62645

6. Which 5 products how the highest percentage of purchases with discount applied?

	Item Purchased text	Discount Rate % numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

7. Segment customers into new returning and loyal Bass on their total number of previous purchase and show the count of each segment?

	Customer Segment text	Number of Customer bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. What are the top 3 most purchased products within each category?

	Category text	Item Purchased text	Total_Orders bigint
1	Accessori...	Jewelry	171
2	Accessori...	Sunglasses	161
3	Accessori...	Belt	161
4	Clothing	Blouse	171
5	Clothing	Pants	171
6	Clothing	Shirt	169
7	Footwear	Sandals	160
8	Footwear	Shoes	150
9	Footwear	Sneakers	145
10	Outerwear	Jacket	163
11	Outerwear	Coat	161

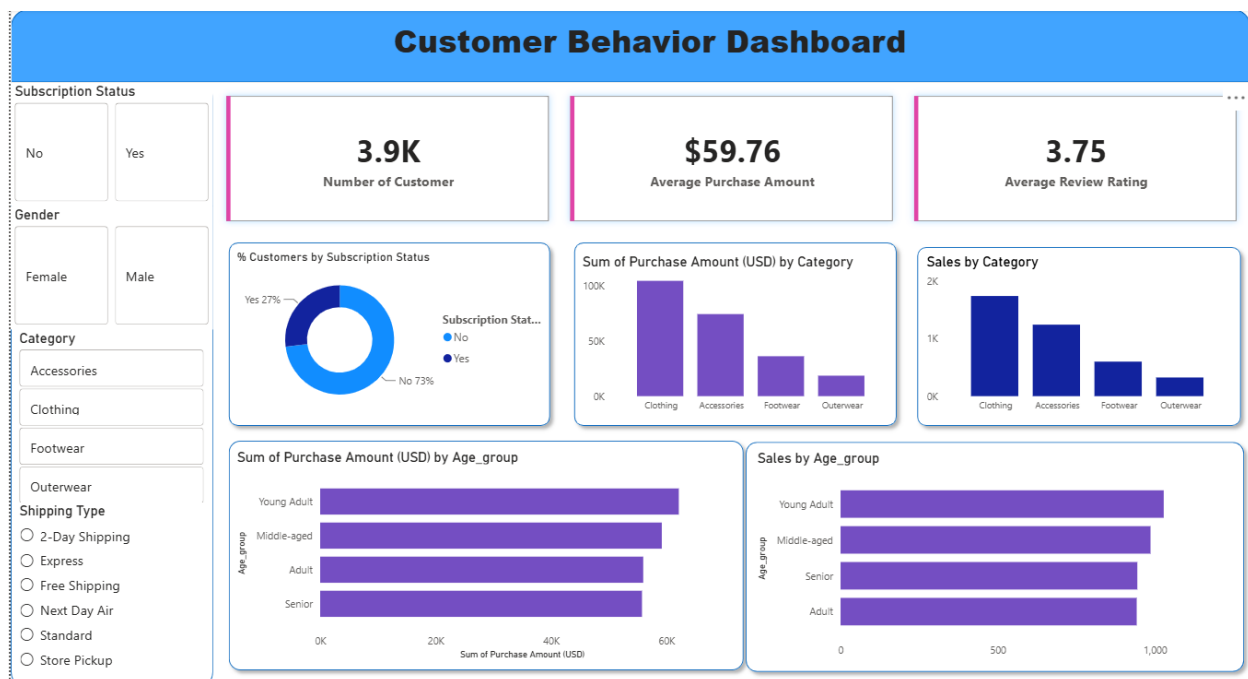
9. Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

	Subscription Status text	Repeat Buyers bigint
1	No	2518
2	Yes	958

10. What is the revenue contribution of each age group?

	Age_group text	Total Revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5.Customer Behavior Dashboard



6.Business Recommendations

- Boost Subscription – Promote exclusive benefits for subscribers.
- Customer Loyalty Programs – Rewards repeat buyers to move them into the ‘Loyal’ Segment.
- Review Discount Policy – Balance sales boosts with margin control.
- Product Positioning – Highlight top-rated and best-selling products in campaigns.

- Targeted Marketing – Focus efforts on high-revenue age group and express-shipping Users.